

Jared T. Nash

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Education	Brown University Sc.M. Physics	2024 - <i>present</i>
	Albion College B.A. Physics, Mathematics <i>Honors Graduate</i>	2019 - 2023
Awards and Honors	ETS Walton Award for Most Outstanding Senior Physics Major, Albion College ('23) David S. Seeley Award for Most Outstanding Junior Physics Major, Albion College ('22) Clement Eugene Rood and L. Katharine Rood Lyday Endowed Physics Scholarship, Albion College ('21) Albion College Fellow and Dean's List (2019-2023)	
Skills	Programming Languages: Wolfram Mathematica, Python (numpy, scipy, matplotlib, pandas), Julia, HTML Operating Systems: Windows, Unix Software: LaTeX, Ansys HFSS, SolidWorks, Excel	
Research Experience	Brown University Mentored by Jonathan Pober (Brown University), The Pober group focuses on developing new techniques for studying the Epoch of Reionization and associated Cosmic Dawn. My role in the Pober group involves modeling radio antenna beam patterns for the Murchison Widefield Array in Australia. In particular, I run experiments on interpolation and emulation methods of beam response models with the goal of producing faster, more reliable methods of simulation. This is done by testing runtime efficiencies of <code>pyuvdata</code> response methods, with future tests to be conducted on the Model for Emulating Directivities and Electric fields of Antennas (MEDEA).	August 2025 - <i>present</i>
	University of Maryland - SSTPRS Mentored by James S. Gates Jr. (UMD), Konstantinos Koutrolikos (UMD), The research of Prof. Gates' group focuses on mathematical models where the property of supersymmetry is investigated in fundamental areas of mathematics and theoretical physics, principally within the domains of particle, field, and string theory. The <i>Summer Student Theoretical Physics Research Session</i> (SSTPRS) is an intensive program focused on developing mathematical techniques for research in particle physics, particularly supersymmetry.	June 2023 - July 2023
	University of Minnesota - REU Mentored by Shaul Hannany (UMN), The Hanany Lab builds instruments to observe the cosmic microwave background radiation (CMB). As a participant in Research Experience for Undergraduates (REU), I conducted computational electromagnetic (CEM) wave simulations to optimize sensor performance for future balloon-borne experiments. In particular, measuring the performance of laser ablated subwavelength structures on alumina lenses using Ansys	June 2022 - August 2022

HFSS with the goal of measuring transmittance and reflectance. This was done in partnership with the Minnesota Super-Computing Institute.

Albion College

September 2021 - May 2022

Mentored by Shahid Iqbal (WMU),

Dr. Iqbal's research involves synthesis of metallic nanostructures, plasmonic, ion beam materials analysis, characterization of nanostructures, and the interaction between quantum emitters and nano-antennas. My role involved the spectrographic analyses of luminescent and fluorescent dyes (Coumarin, BP Fluor 350) to test the effects of the presence of silver nanoparticles.

**Teaching
Experience**

The Wheeler School

October 2024 - *present*

Aerie Instructor, Providence, RI

- Develop and implement engaging lesson plans to effectively deliver educational content in areas of physics and mathematics, most notably quantum mechanics
- Head of after-school elementary math instruction and engagement
- Lead small-group tutoring sessions targeting foundational math skills
- Adopt methods and materials to student's varying needs and interests

Achievement First Mayoral Academy

July 2023 - July 2024

Mathematics Instructor, Cranston, RI

- Used data-driven methods to Evaluate 150+ students' progress via weekly assessments, improving class average by over 20% on Rhode Island State Assessments
- Tracked growth for struggling learners and designed targeted intervention plans
- Motivated students in science and technology related projects and experiments

Albion College

August 2022 - May 2023

Teaching Assistant, Albion, MI

- Evaluated homework, quizzes, and exams with detailed feedback on reasoning and technique
- Supervised introductory physics laboratory sections, guiding students in data collection and analysis

**Talks and
Posters**

Pober et al. (2025) *Radio Cosmology Lab, Brown University Physics Department*
Link: [here](#)

Nash et al. (2023) *The Anti-reflective Properties of Subwavelength Structures on Alumina Lenses*, Albion College Elkin R. Isaac Student Research Symposium. Oral presentation.

Nash et al. (2022) *The Anti-reflective Properties of Subwavelength Structures on Alumina Lenses*, University of Minnesota Department of Physics REU. Oral presentation and Poster
Link: [here](#)

Outreach

Science Olympiad Coach and Mentor - The Wheeler School, Providence RI.

- Coach and mentor students for national and state Science Olympiad competitions. The Science Olympiad ranges in subject from robotics, to fossil analysis, to astronomy.

Albion College Astronomy Club (2021 - 2023)

- Lead public outreach and educational events, including an open telescope night for the public.

Coursework

- Differential Equations
- Partial Differential Equations
- Linear Algebra
- Abstract Algebra
- Number Theory/Discrete Mathematics
- Real/Complex Analysis
- Theoretical Mechanics
- Quantum Mechanics I, II
- Electrodynamics
- Astrophysics
- General Relativity
- Thermodynamics and Statistical Mechanics